



Charlie (Zhaohua) Qi

Carrum Downs, VIC 3201 | +61 451 176 394

charlieqi2017@gmail.com | github.com/Charlie-Qi394

Portfolio: charlie-qi394.github.io/charlie-qi-portfolio

PROFESSIONAL SUMMARY

Master of Computer Science graduate specialising in Artificial Intelligence, targeting full-stack software engineering, Python, data and applied AI/ML roles. Built production-style projects spanning nutrition formulation optimisation, AI document extraction, MCP-powered operations, RAG, CNN classification and semantic segmentation using TensorFlow/Keras and PyTorch. Brings 10 years of experience across biotech and nutrition product R&D, infant formula manufacturing and healthcare nutrition, translating complex domain workflows into auditable software, data models and decision tools. Demonstrates learning agility through first-in-class results in three computer science units and a High Distinction average.

TECHNICAL SKILLS

Full-Stack Software Engineering

Python, TypeScript, React, FastAPI, Node.js, Express, REST APIs, Celery, Redis, pytest/Vitest

Data / Optimisation

PostgreSQL, SQLAlchemy, SciPy/HiGHS, SQL, pandas, NumPy, mass-balance modelling

AI / Extraction / RAG

Gemini structured extraction, MCP tools, provider APIs, pgvector, vector search, LangGraph

DevOps / Automation

Docker Compose, Caddy, GitHub Actions, audit logging, VBA tools, SAP/M3, QMS/PLM/LIMS/ELN exposure

Deep Learning / CV

PyTorch, TensorFlow/Keras, CNNs, U-Net/FPN, attention gates, segmentation, mIoU

Security

RSA, AES, X.509, CSRs, CRLs, PKI concepts

SOFTWARE, DATA AND AI PROJECTS

Integrated Nutrition Formulation Optimisation Platform

[GitHub](#)

React, TypeScript, FastAPI, PostgreSQL, SciPy/HiGHS, Gemini

- Designed and built a production-style platform that ingests supplier Excel, CSV and digital PDF nutrient evidence, requires formulator review, and publishes immutable ingredient, specification and optimisation versions with audit history.
- Implemented multi-stage linear optimisation for macronutrient and formulation-ratio targets, natural micronutrient mass balance, base-powder premix gap calculation, supplier-proposal reconciliation and deterministic specification-margin risk bands.

CareOps AI - MCP-Powered Aged Care Operations Assistant

[GitHub](#)

TypeScript, Node.js, MCP, PostgreSQL

- Built a production-style AI operations platform for a fictional Australian aged-care provider using TypeScript, Express, React, PostgreSQL, Prisma, Docker and GitHub Actions.
- Implemented a real MCP server with secure tools for client search, worker availability, compliance checks, address updates and appointment rescheduling, including RBAC, Zod validation, confirmation-gated writes and audit logging.

AI Regulatory Knowledge Assistant

[GitHub](#)

RAG, FastAPI, PostgreSQL/pgvector

- Built a document-grounded RAG assistant using FastAPI, Streamlit, PostgreSQL/pgvector, OpenAI embeddings and chat generation to answer questions over local regulatory documents with citations and query history.

Computer Vision Algorithms and Deep Learning

[GitHub](#)

TensorFlow/Keras, CNNs, semantic segmentation

- Implemented Harris corner detection and a Canny edge-detection pipeline; built and compared 12 CNN classification variants, reaching 74.18% final tuned test accuracy.
- Designed and evaluated FCN, U-Net and FPN/ASPP/attention segmentation models, improving validation mIoU from 0.5332 to 0.6849 within a <15M-parameter constraint.

Python PKI Certificate System

[GitHub](#)

Python, cryptography, X.509

- Implemented an educational PKI simulator with Root CA, Sub-CAs, clients, encrypted certificate requests, validation and revocation.

Seq2Seq Recipe Generation with Attention

[GitHub](#)

NLP, PyTorch

- Built a PyTorch LSTM encoder-decoder recipe-generation system using ingredient lists as source sequences and recipe instructions as targets.

Charlie (Zhaohua) Qi

Biotech product R&D, manufacturing, healthcare and transferable systems experience

BIOTECH PRODUCT R&D, MANUFACTURING AND SYSTEMS

Technical Manager - Research and Development in Infant Formula

Melbourne | Apr 2022 - Present

Bellamy's Organic

- Lead infant formula and nutritional product development from concept to launch, coordinating marketing, sales, suppliers, regulatory stakeholders and third-party manufacturers across the product lifecycle.
- Manage formulation assessment, nutrition panels, ingredient sourcing, packaging validation, sensory evaluation, shelf-life validation, specification records and controlled technical documentation in QMS/PLM-style workflows.
- Support regulatory and market-access work including SAMR GB-2 registration, China organic WIT certification and FDA exemption-related technical evidence.
- Use SAP and cross-functional R&D, QC and manufacturing data inputs to support technical decisions, supplier coordination, product changes, traceability and operational follow-up.
- Developed an HTML ingredient forecasting calculator to support sales and procurement planning, improving business visibility and stakeholder alignment.

Formulation Technologist - Operation

Jan 2020 - Apr 2022

Fonterra Australia

- Supported new product/process development, drying and processing challenge resolution, equipment commissioning and campaign-management improvement in a manufacturing environment.
- Used SAP, production records and laboratory/testing data to support formulation, scheduling, QC follow-up, process traceability and operational decision-making across manufacturing campaigns.
- Developed a production calculator to automate ingredient conversion calculations, reducing manual work and saving operational time.

Product Development Technologist - Nutritionals

Apr 2018 - Jan 2020

Bega Cheese

- Developed formulations, bills of materials, specifications, operating conditions and regulatory documentation across nutritional powder categories, supporting PLM-style product data control.
- Used planning and business systems including M3 to support product development, formulation changes, specification updates, manufacturing readiness and cross-functional traceability.
- Built a goat milk formulation calculator in VBA to automate manufacturing operation calculations and improve process consistency.

Dry Blending Formulator and QA Officer

Sep 2017 - Apr 2018

Australian Natural Milk Association

- Supported dry-blending formulation, GMP audits, supplier documentation, QC/QMS records, corrective actions and product-compliance reporting.

HEALTHCARE EXPERIENCE

Clinical Dietitian Placement

2016

Sydney Hospitals

- Completed clinical nutrition placements at Concord Hospital, Sydney Prince Alfred Hospital, Bondi Mental Health Clinic and Sydney Westmead Children's Hospital.
- Used electronic patient record and hospital information systems to review clinical notes, nutrition history and care documentation.
- Supported paediatric/infant nutrition, allergies and intolerances, rehabilitation nutrition, food-safety auditing and clinical research activities.

EDUCATION

- **Master of Computer Science, Artificial Intelligence specialisation** | Monash University | Completed Jun 2026

First in class Machine Learning; Computer Vision; Applied Practice 1.

Result High Distinction average.

- **Master of Science, Nutrition and Dietetics** | The University of Sydney | 2016
- **Bachelor of Science, Food Science and Nutrition** | The University of Auckland | 2014

First in class Food Quality Attribute (highest score).

Award Certificate of Distinction in Statistics for top 10%.

COURSEWORK EVIDENCE

- Relevant coursework: NLP, Machine Learning, AI, Computer Vision, Cryptography, Databases, Software Engineering, Python and Applied Practice.
- Databases and Data Handling: conceptual and logical database design, relational schema design, SQL scripting, data preprocessing and exploratory analysis.
- Systems and Networking: routing, EIGRP metric analysis, static routes, firewall configuration, Wireshark packet analysis and TCP/IP concepts.